COMP90072

The Art of Scientific Computing

Two-dimensional sandpile dynamics

Avalanche statistics and random redistribution

Final coursework report

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Date: 18 October 2026

A numerical study of the Bak–Tang–Wiesenfeld model on finite open grids, with a comparison to independent random redistribution of four grains. The accompanying software contains the saved trajectories, a function-based Python implementation, and the data used to produce the figures.

Abstract

A local threshold rule can turn a one-grain input into an avalanche involving many sites. I simulated this behaviour in the two-dimensional Bak–Tang–Wiesenfeld (BTW) model on open grids of width 16, 32 and 64. Each setting used three runs with 20,000 recorded additions per run after burn-in. The mean grid height increased from 2.0465 to 2.1040 across these sizes, and the mean run-level 99th percentile of nonzero toppling counts increased from 192.66 to 4,329.51. A second rule let each of the four outgoing grains choose its direction independently. At width 32, this reduced the average avalanche area from 78.294 to 50.042 sites while increasing the conditional mean number of topplings from 96.705 to 103.688. Both rules retained an all-addition mean loss close to one. Exact mass checks and independent update-order tests supported the implementation. The measured tail shapes remained sensitive to system size and fitting threshold, so the fitted exponents are treated as exploratory. The results show how the same average transport rule can accompany noticeably different avalanche geometry and event frequencies.

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Notation. L is the grid width; s the number of topplings; a the number of distinct toppled sites; d the boundary loss; r the maximum propagation radius; and H the mean stable height. Unless a caption says otherwise, error bars and "±" values describe the sample standard deviation of three run-level estimates.

1 Introduction

A sandpile provides a simple setting for a question about collective behaviour: how much can a system change after one small input? In this project, every input is one grain, yet the resulting number of topplings can vary greatly. I followed the model and questions in the supplied course handout, including its option to randomise the direction of each outgoing grain [1].

The BTW model was introduced as an example of self-organised criticality [2]. It combines slow external driving with rapid local relaxation. A site responds only when its height reaches a threshold, and its response can push neighbouring sites over their thresholds. This gives a route from a small local change to a much larger event. The separation of timescales is imposed directly here: another grain is added only after the preceding avalanche has ended.

1.1 Questions and scope

The first aim was to implement the deterministic model and check that its results followed the stated rules. I then examined the approach to a stationary regime, four measures of avalanche size, their correlations, and the effect of grid width. The extension asks how these measurements change when redistribution becomes random while the threshold, drive and open boundary remain fixed.

The numerical study uses $L = 16, 32, 64$ for BTW and $L = 32$ for the extension. This scale is manageable with a short Python program and allows repeated runs and diagnostic checks. I expected the height to rise from an empty grid and eventually fluctuate within a narrower range. An initial guess of 1.5 came from assuming equal frequencies of the four stable heights. That assumption is tested explicitly in Chapter 4. For the extension, the preserved input and average outgoing transfer suggested similar long-run mass balance. The direction of the changes in avalanche geometry was left to the comparison.

1.2 Scale invariance and the finite-grid setting

Suppose a nonzero statistical function retains its shape under a continuous range of rescalings. A homogeneous representation is

$$f(bx) = b^{-\tau} f(x). \tag{1.1}$$

For differentiable f, differentiating with respect to b at $b = 1$ gives $xf'(x) = -\tau f(x)$. Integration yields $f(x) = Cx^{-\tau}$. A power law therefore has no characteristic scale within the range where this relation holds. Discrete rescaling alone is weaker and may allow log-periodic corrections. For a mass function with tail $p(s) \propto s^{-\tau}$ and $\tau > 1$, the corresponding far-tail complementary cumulative distribution has exponent $1 - \tau$ under the usual tail approximation.

The lattice spacing sets a minimum length, while the finite grid bounds the area and propagation radius. Any apparent scaling must be assessed within those limits. I use distribution curves and several lower fitting thresholds to see how far a simple power-law description can be taken.

1.3 Why the boundary matters

A finite closed grid gains one grain at every driving step. Stable configurations hold at most $3L^2$ grains, so continued conservative driving eventually prevents complete stabilisation. Open boundaries provide an outlet. An infinite grid requires a different specification: a uniform positive selection probability cannot be assigned to every site of $\mathbb{Z}^2$. Driving a finite subset is possible, but outward transport from a finite observation window leaves grains elsewhere in the infinite system. Statements about its limiting state also require an initial configuration and a defined order of limits. I therefore use the finite open model for the numerical work.

2 Model and stabilisation theory

2.1 Local rule and recorded quantities

The state is an $L \times L$ array of nonnegative integer heights, initially zero. A uniformly selected site receives one grain. Whenever $h(i, j) \geq 4$, that site loses four grains and sends one in each of the four nearest-neighbour directions. A grain directed outside the grid is removed. This is repeated until all heights are below four. Edge and corner sites retain the same threshold. A driving step counts one addition and its complete relaxation.

If $u(x)$ is the number of topplings at site x during that relaxation and x_0 is the addition site, the recorded quantities are

$$s = \sum_x u(x), \qquad a = \#\{x : u(x) > 0\}, \qquad r = \max_{u(x)>0} \|x - x_0\|_2, \qquad H = \frac{1}{L^2} \sum_x h(x). \tag{2.1}$$

Boundary loss d counts grains that leave during the same relaxation. Lattice spacing is one. The radius directly measures how far the disturbance reaches and is inexpensive to calculate from the toppled sites. A step without a toppling has $s = a = d = r = 0$. A one-site avalanche also has $r = 0$, so the event definition is always $s > 0$. Repeated topplings increase s while contributing only once to a.

2.2 Toppling matrix and dissipation

The Abelian formulation gives a useful way to express the update and study its dependence on processing order [3]. I use the handout's negative-diagonal convention: row x gives the increment caused by toppling at x,

$$\Delta(x, y) = \begin{cases} -4, & x = y, \\ 1, & x, y \text{ are nearest neighbours within the grid,} \\ 0, & \text{otherwise.} \end{cases} \tag{2.2}$$

Its conditions have direct interpretations (Table 1). Symmetry describes the undirected transfer in this model; directed dissipative sandpiles can be defined as well.

Table 1: Meaning of the toppling-matrix conditions.

Condition	Meaning
$\Delta(x, y) = \Delta(y, x) \geq 0, x \neq y$	A toppling supplies nonnegative amounts to other sites, with symmetric transfer.
$\Delta(x, x) < 0$	It removes grains from its own site; the threshold is $-\Delta(x, x)$.
$\sum_y \Delta(x, y) \leq 0$	It conserves or reduces total mass.
$\sum_{x,y} \Delta(x, y) < 0$	At least one site is dissipative.

For an interior site the row sum is zero; at a non-corner edge it is -1, and at a corner it is -2. The total is $-4L$ for $L \geq 2$. A single-site grid has matrix $[-4]$. With this sign convention, the instability condition is $h(x) \geq -\Delta(x, x) = 4$; the printed sign in the handout's Equation (6) is inconsistent with its stated rule.

Every site must also have a path to a dissipative site. The four conditions alone do not guarantee this on a disconnected graph. For example,

$$\Delta = \begin{pmatrix} -1 & 1 & 0 \\ 1 & -1 & 0 \\ 0 & 0 & -1 \end{pmatrix} \tag{2.3}$$

satisfies them, but a grain at site 1 can move indefinitely between sites 1 and 2 under the threshold-one rule. The open square grid has the required accessibility.

2.3 Why legal relaxation has a unique outcome

The following argument separates termination from uniqueness. It follows the finite-graph treatment and least-action reasoning described by Járai [4]. During relaxation the initial mass is finite, no new grains arrive, and all heights remain nonnegative.

2.3.1 Termination

Suppose a legal relaxation contains infinitely many topplings. Since the grid has finitely many sites, at least one site topples infinitely often. Each of its neighbours then receives infinitely many grains. A neighbour that toppled only finitely often would accumulate unbounded height, contradicting the finite upper bound on total mass. Thus every neighbour must also topple infinitely often. Following a path through the connected grid reaches a boundary site. Its repeated topplings would lose infinitely many grains, again contradicting the finite initial mass. The relaxation must end.

2.3.2 Exchanging two legal topplings

Let Δ_x be row x viewed as an increment vector. An executed toppling gives $T_x h = h + \Delta_x$. If two sites x and y are both unstable, toppling either cannot reduce the height of the other. Both orders remain legal, and

$$T_y T_x h = h + \Delta_x + \Delta_y = T_x T_y h. \tag{2.4}$$

The legality condition matters for a conditional operation that does nothing at a stable site. Starting from $\left(\begin{smallmatrix} 3 & 4 \\ 0 & 0 \end{smallmatrix}\right)$, select the upper-left site x and upper-right site y. Applying T_x then T_y gives $\left(\begin{smallmatrix} 4 & 0 \\ 0 & 1 \end{smallmatrix}\right)$; applying T_y then T_x gives $\left(\begin{smallmatrix} 0 & 1 \\ 1 & 1 \end{smallmatrix}\right)$. In the first order, x still needs to topple. Completing the relaxation gives the same final state. This resolves the ambiguity in treating every conditional operation as freely interchangeable.

2.3.3 Uniqueness of the complete relaxation

Take two complete legal stabilisations from the same initial state h, with total site-level toppling counts $u(x)$ and $v(x)$. Follow the first sequence and suppose a site's count is about to exceed v for the first time. Just before that toppling at x, the partial counts n satisfy $n(x) = v(x)$ and $n(y) \le v(y)$ for every other site. The current height at x is therefore

$$\begin{aligned} h(x) + \Delta(x,x)n(x) + \sum_{y \neq x} \Delta(y,x)n(y) &\le h(x) + \Delta(x,x)v(x) + \sum_{y \neq x} \Delta(y,x)v(y) \\ &= h_{\text{final},v}(x) < 4. \end{aligned} \tag{2.5}$$

The inequality uses the nonnegative off-diagonal entries. The proposed toppling would be illegal, which contradicts its selection. Hence $u \le v$. Reversing the two sequences gives $v \le u$, so $u = v$ and

$$h_{\text{final}} = h + \Delta^{\mathsf{T}} u \tag{2.6}$$

is unique. This establishes both the final heights and the number of topplings at every site. It justifies choosing a convenient legal processing order in the program.

The associated addition-and-stabilisation chain, with positive driving probability at every site, has a single recurrent communicating class. Its stationary measure is uniform on recurrent configurations. On that class, the process is ergodic for time averages under the stated driving conditions. The time and ensemble comparison in Section 4.2 uses this setting. Configurations generated by independently selecting heights from $0, 1, 2, 3$ form a different sample space.

3 Computational methods

3.1 A candidate-list relaxation algorithm

The core program uses NumPy integer arrays and a double-ended queue. Each drive begins by adding a grain and placing its site in the candidate list. The last candidate is removed first. Its height is checked again, since a listed site need not still require a toppling. A legal toppling updates four neighbours, the site-level count array, and the boundary loss. A neighbour is added when it first reaches height four. The source is added again if it remains unstable. Figure 1 shows the complete loop.

This arrangement avoids scanning the entire grid to choose each next toppling. The local work is proportional to the number of processed candidates and four transfers per executed toppling. Computing the recorded area, radius and full-grid checks adds array work of order L^2 per addition. That cost is acceptable for the selected sizes and keeps the implementation easy to inspect. Only one height grid and the current toppling-count array are needed during relaxation; the saved trajectory stores five scalar measurements per driving step.

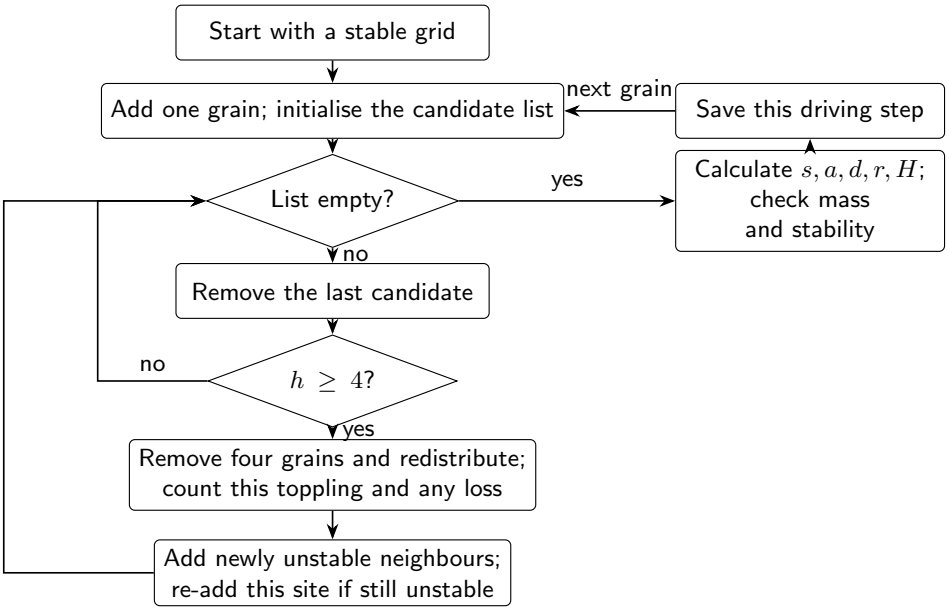


Figure 1: Candidate-list algorithm for one grain addition and complete relaxation. The same loop handles both rules; only the choice of outgoing directions changes. A new driving step starts after the candidate list is empty and the checks have passed.

3.2 Checks before collecting statistics

For every simulated step the program checks the exact integer identity

$$M_{t+1} = M_t + 1 - d_t, \tag{3.1}$$

as well as nonnegative stable heights and $a \leq s$. Local tests cover centre, edge and corner topplings. A separate synchronous array algorithm provides a cross-check on the candidate-list implementation. All 1,024 additions to stable 2×2 grids, together with 100 random unstable 8×8 configurations, produced identical final heights, site-level toppling counts and loss under last-in-first-out, first-in-first-out and synchronous processing.

The 1,124 comparisons passed again during report preparation. A read-only audit of all 782,220 saved steps found zero mass-balance residual. These tests support the implementation; the complete mathematical statement about processing order is established in Section 2.3.

3.3 Sampling design and statistical summaries

Each run starts empty. The initial burn-in is $B = \max(5000, 10L^2)$, followed by 20,000 recorded additions. Three seeds are used per main setting. Burn-in is checked through height windows, comparisons of the two halves of the recorded phase, and an additional doubled-burn-in experiment at $L = 32$ (Table 2).

Table 2: Experiments included in the saved data. Recorded additions exclude burn-in.

Experiment	L	Burn-in	Recorded	Runs
BTW	16	5,000	20,000	3
BTW	32	10,240	20,000	3
BTW	64	40,960	20,000	3
Random redistribution	32	10,240	20,000	3
Doubled burn-in, each rule	32	20,480	20,000	3 each
Independent BTW ensemble	16	5,000	1	20

The main experiments contain 240,000 recorded additions and 99,143 nonzero avalanches. Summary means and quantiles are calculated within each run before taking the mean and sample standard deviation across the three runs. Distribution curves pool the three equal-length records. Spearman correlations use all events with $s > 0$, including events with zero loss or radius.

For logarithmic plots I use the exact positive-value CCDF, $P(X \geq x \mid X > 0)$, separately for each variable. Zero frequencies remain in the full data. The exploratory fitting calculation follows the discrete likelihood formulation of Clauset and colleagues [5]:

$$p(s \mid s \geq s_{\min}) = \frac{s^{-\alpha}}{\zeta(\alpha, s_{\min})}, \qquad -\ell(\alpha) = n \log \zeta(\alpha, s_{\min}) + \alpha \sum_{i=1}^{n} \log s_i. \tag{3.2}$$

I minimise this expression for $\alpha > 1$ at six fixed thresholds. The output contains the tail sample size and the maximum empirical-to-fitted CDF distance (KS distance). No bootstrap goodness-of-fit test or comparison with alternative distributions was performed. The site-level count array also gives a spatial check on the avalanche measurements (Figure 2).

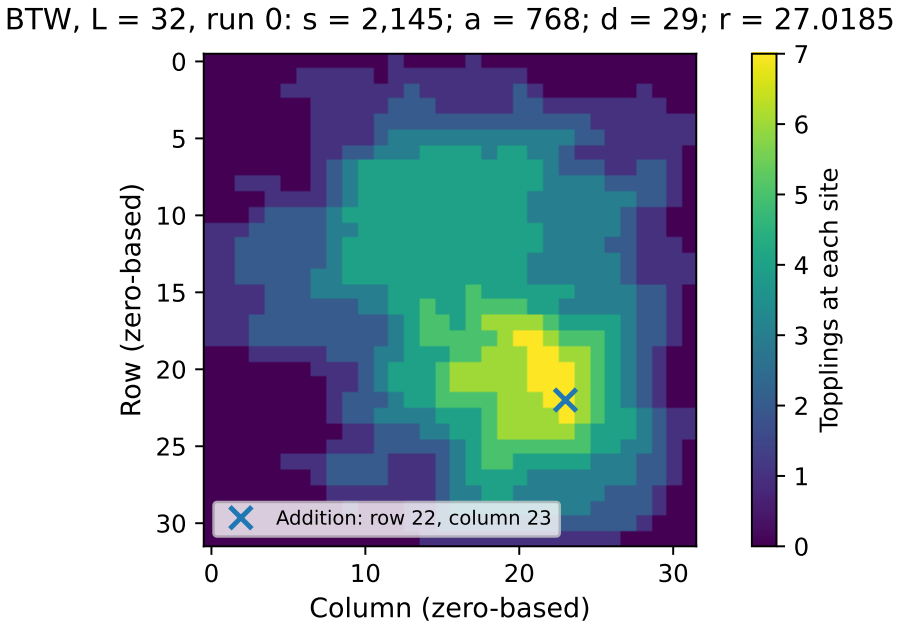


Figure 2: Site-level topplings for the largest recorded avalanche in BTW $L = 32$, run 0. It occurs at post-burn-in addition 13,646 and has $s = 2{,}145$, $a = 768$, $d = 29$ and $r = 27.0185$. The cross marks zero-based row 22, column 23; a site topples at most seven times. This is a selected large event from one run.

4 Results for the BTW model

4.1 Loading, mean height and the stationary regime

The initial height guess was $(0 + 1 + 2 + 3)/4 = 1.5$. It assumes equal frequencies of the four stable heights and ignores the dependence created by redistribution. The measured averages are all higher. Figure 3 shows the early loading and later fluctuations for one run; Figure 4 compares the averages across all three runs per setting.

For the infinite square lattice, the exact stationary BTW mean is $17/8 = 2.125$ under the present 0–3 height convention. The result is commonly written as $25/8$ with heights 1–4 [6]. The finite-grid averages in Figure 4 increase from 2.0465 at $L = 16$ to 2.1040 at $L = 64$, moving towards that benchmark. This is consistent with the decreasing relative importance of boundary sites.

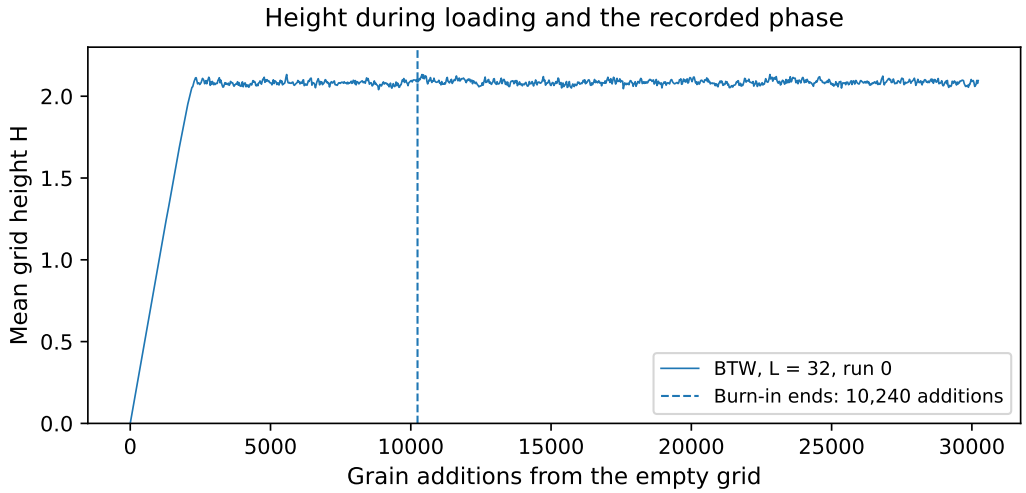


Figure 3: Mean height in BTW $L = 32$, run 0, from the empty grid through the recorded phase. The vertical line marks the end of 10,240 burn-in additions. Every 25th step is displayed; all steps are used in the calculations. This single trajectory is distinct from the three-run summaries below.

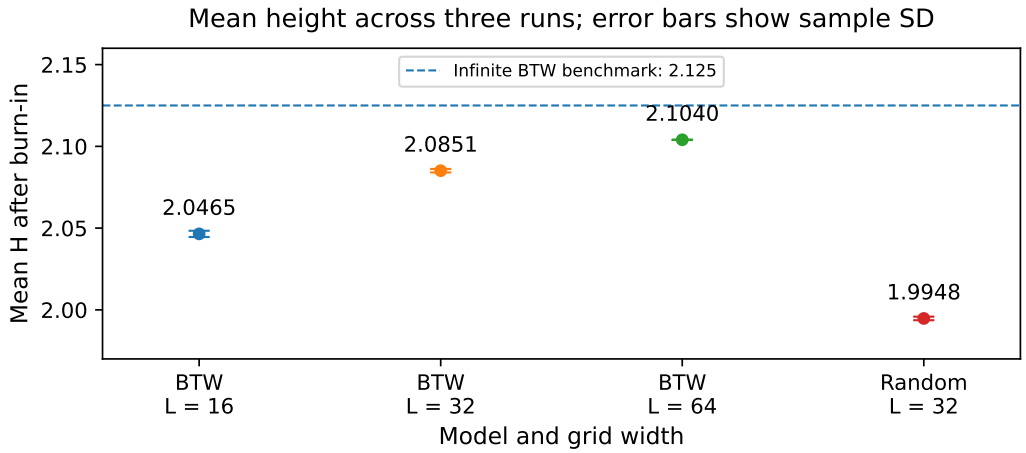


Figure 4: Mean post-burn-in height across three runs per setting. The labels are rounded to four decimal places; error bars show run-to-run sample SD. The random-rule value is included for the later comparison. The dashed benchmark applies to infinite-lattice BTW.

A stable configuration means $h < 4$ everywhere. Statistical stationarity concerns distributions that remain unchanged under a shift of observation time. The late trajectory still changes from one drive to the next, while its height stays in a much narrower range. Here steady state refers to long-run statistics, while thermodynamic equilibrium excludes this sustained driving-and-loss process. Grains continue to enter and leave the system. The height plateau is a useful diagnostic, with the burn-in sensitivity in Section 5.3 providing a further check.

4.2 Time sampling and independent ensembles

Time sampling follows successive states of one system, so neighbouring observations can be correlated. Ensemble sampling draws observations from separately driven systems. The two estimates need sufficient sampling of the relevant recurrent states to be comparable. A short run, an initial transient or strong time correlation can create visible differences.

For BTW at $L = 16$, I took the 20,000 post-burn-in heights from run 0 and divided them into 40 non-overlapping blocks of 500 steps. Their mean is 2.044285. I estimated a standard error from the block means, treating them as approximately independent. The comparison ensemble consists of 20 independently seeded systems, each observed after 5,001 additions from an empty grid. Its mean is 2.055664.

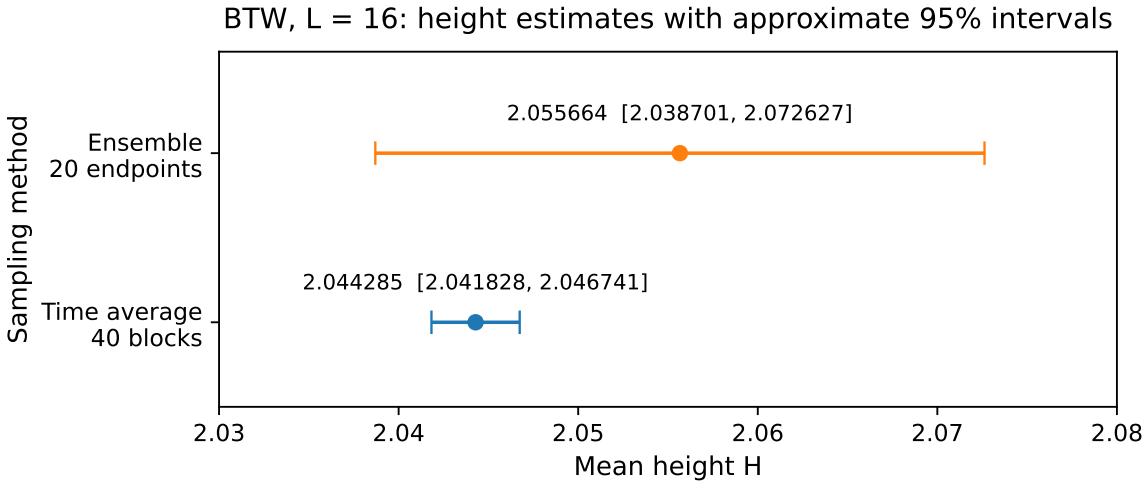


Figure 5: Time and ensemble estimates for mean height at $L = 16$. Horizontal intervals use Student-t multipliers. The time interval treats 40 block means as approximately independent; the ensemble interval uses 20 independent endpoints. Numerical values are also given in Table 3.

Table 3: Values used in Figure 5. These are approximate confidence intervals, unlike the three-run SD bars used elsewhere.

Estimate	Mean H	Standard error	Approximate 95% interval
Time: 40 block means	2.044285	0.001215	[2.041828, 2.046741]
Ensemble: 20 endpoints	2.055664	0.008105	[2.038701, 2.072627]

The time estimate lies within the much wider ensemble interval (Figure 5). The difference is compatible with the sampling uncertainty of these 20 endpoints. The check concerns one observable at one grid width; it does not establish equal sampling accuracy for the avalanche tail.

The theoretical setting in Section 2.3 supports relating long-run time averages to the stationary ensemble on the recurrent class. It does not give uniform sampling over every stable configuration. A burning test in the saved analysis identified all 32 deterministic final configurations as recurrent. This confirms their class membership, while mixing within that class remains a separate issue.

Periodicity also deserves care. On a one-site grid, successive heights follow $1, 2, 3, 0$ repeatedly. Its long-run mean is well defined, but its distribution at a fixed time from a fixed starting state continues to cycle. The numerical test includes this small example. For the larger runs, the block and burn-in diagnostics provide a practical assessment of the available stationary-window estimates.

4.3 Avalanche distributions and grid width

4.3.1 Topplings and area

Both toppling count and area have broad, right-skewed distributions (Figures 6 and 7). Larger grids extend the range of large events. The average run-level 99th percentile of s, conditional on an avalanche, is 192.66, 942.57 and 4,329.51 at $L = 16, 32, 64$ respectively (Table 4). These quantiles are calculated within runs; the plotted CCDFs pool the runs.

Table 4: Main BTW size summaries. Entries are means ± sample SD of three run-level estimates. The last column averages the three separately calculated event quantiles.

Setting	$P(s > 0)$, percent	Mean s, all additions	99th percentile, $s > 0$
BTW $L = 16$	40.982 ± 0.319	11.3331 ± 0.0491	192.66 ± 6.04
BTW $L = 32$	41.883 ± 0.201	40.5030 ± 0.2150	942.57 ± 11.17
BTW $L = 64$	43.175 ± 0.157	153.1406 ± 0.4993	4329.51 ± 153.25

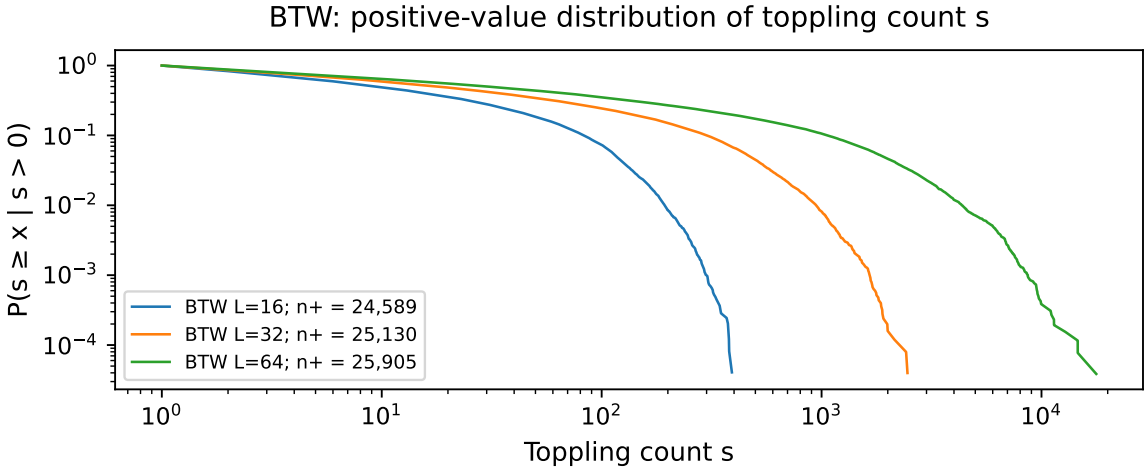


Figure 6: Positive-value CCDF of the number of topplings. Each curve pools three runs, each with 20,000 recorded additions. The legend gives the number of nonzero observations. The larger-L curves extend to substantially greater toppling counts.

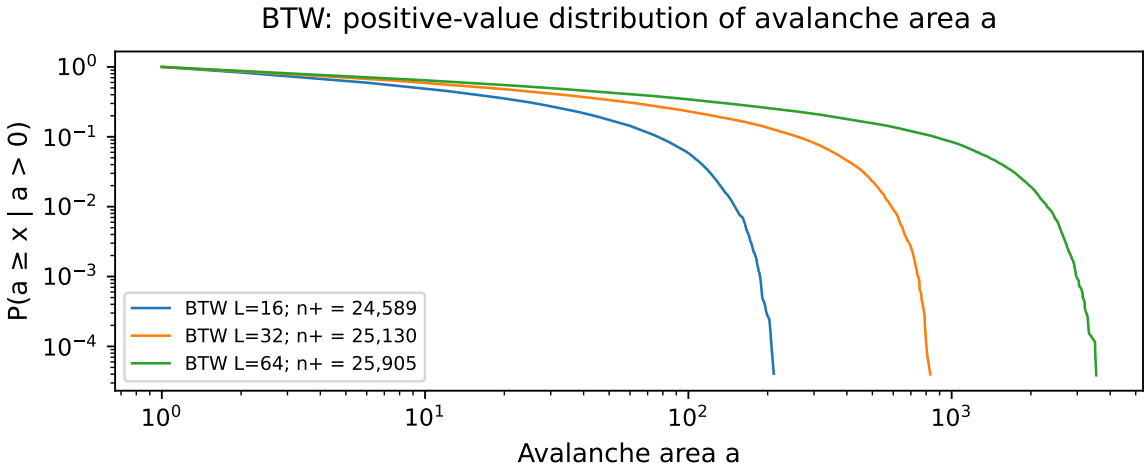


Figure 7: Positive-value CCDF of avalanche area for the same runs. The positive sample counts equal those for s, because an avalanche has at least one toppled site. Area is bounded by L^2; repeated topplings do not add new sites.

The similar ordering of these curves is expected because avalanches with many topplings often visit more sites. Their numerical values still differ. For example, the event in Figure 2 has 2,145 topplings spread over 768 sites.

4.3.2 Boundary loss and propagation radius

Loss is zero for many additions, including some that trigger avalanches. Radius can also vanish during a one-site avalanche. Table 5 gives these zero frequencies. Figures 8 and 9 show only the positive part of each variable, so they have different sample sizes and conditional denominators.

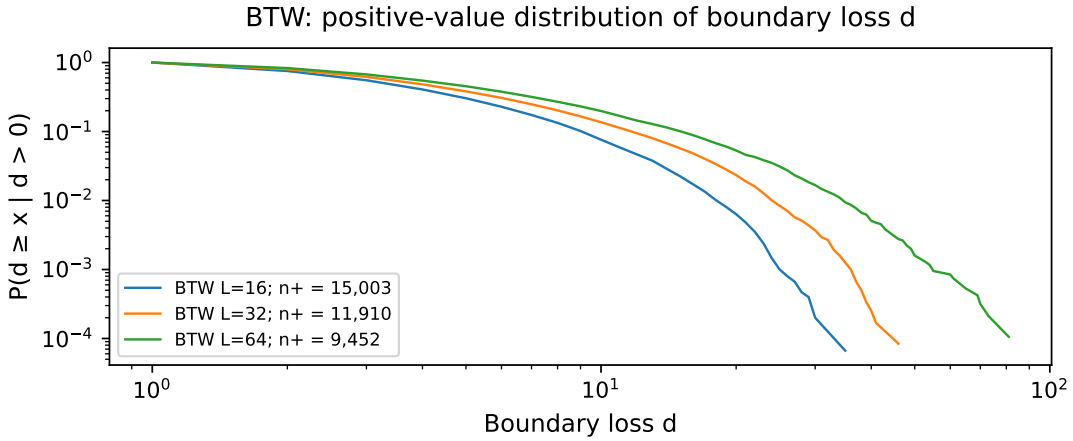


Figure 8: Boundary loss conditional on $d > 0$. Larger grids have fewer loss-producing additions in these records (Table 5), although positive loss events can be larger. Mean loss over all additions stays close to one.

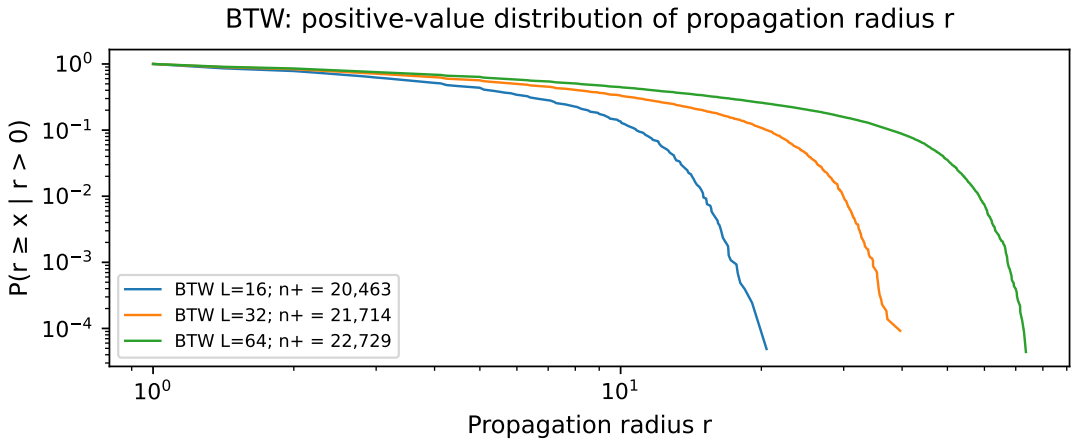


Figure 9: Propagation radius conditional on $r > 0$. Distances are Euclidean with unit lattice spacing. The bound is $r \leq \sqrt{2}(L-1)$. The outward shift with L reflects the larger available spatial extent.

Table 5: Zero-valued additions, as percentages of all 60,000 recorded additions per setting. The random setting is included for Chapter 5.

Model	$s = 0$	$a = 0$	$d = 0$	$r = 0$
BTW $L = 16$	59.018	59.018	74.995	65.895
BTW $L = 32$	58.117	58.117	80.150	63.810
BTW $L = 64$	56.825	56.825	84.247	62.118
Random $L = 32$	60.802	60.802	84.793	65.622

The positive loss and radius distributions are also right-skewed. Loss includes a substantial probability at zero, while radius is restricted to distances available on the lattice. A symmetric Gaussian would miss these features. I applied the exploratory power-law fit to toppling count, whose discrete tail can be specified directly.

4.4 Relationships between the measurements

Before calculating the correlations, I expected toppling count, area and radius to be positively related. An avalanche that keeps spreading usually both visits more sites and performs more topplings. Loss should have a less direct relationship because the location of the initial addition and contact with the boundary also matter.

The measurements support this expectation. In BTW at $L = 32$, the Spearman correlation of s with a is 0.9992, and that of s with r is 0.9836 (Figure 10). The correlation of s with loss is lower, at 0.5532. Table 6 shows every pair for this grid width and the random extension.

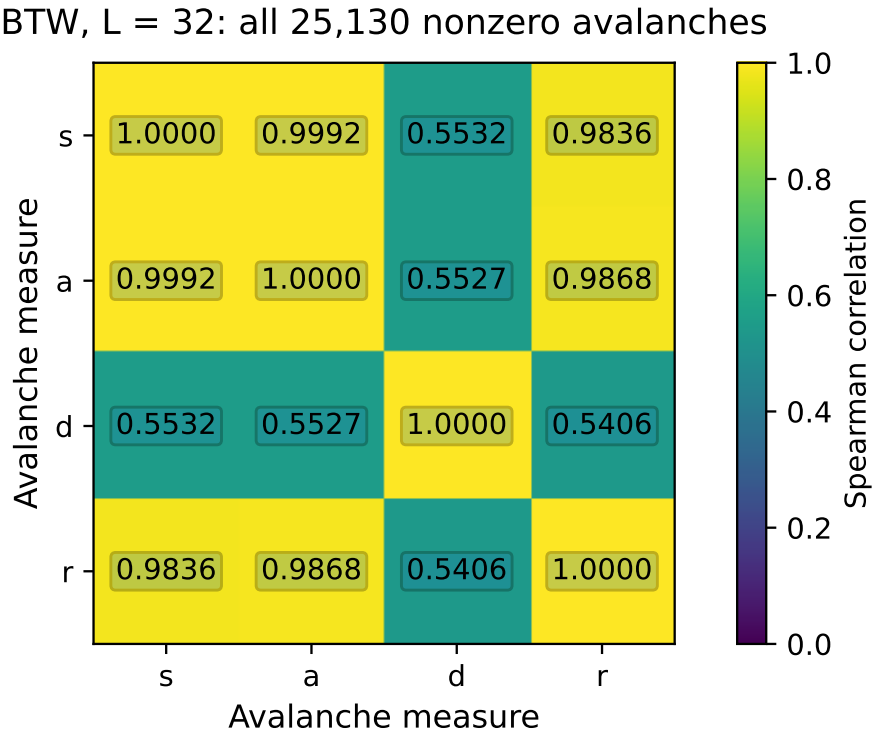


Figure 10: Spearman correlations among all 25,130 nonzero BTW avalanches at $L = 32$, pooled across three runs. Zero loss and zero radius are retained. Displayed values use four decimal places and match the BTW column of Table 6.

Table 6: Descriptive rank correlations at $L = 32$. The random model contributes 23,519 nonzero avalanches.

Pair	BTW	Random redistribution
s, a	0.9992	0.9986
s, d	0.5532	0.4949
s, r	0.9836	0.9798
a, d	0.5527	0.4942
a, r	0.9868	0.9835
d, r	0.5406	0.4827

A high rank correlation describes the ordering of observations. It allows a substantial difference in their magnitudes: $a \leq s$, and repeated topplings can make the gap large. This becomes particularly relevant to the random extension. The weaker loss correlations fit the additional role of the boundary, although the correlation matrix alone cannot separate the contributions of initial position and avalanche geometry.

The estimates use every nonzero event in the chosen records. Since events are generated along trajectories, I use these coefficients descriptively and do not attach independence-based significance tests. The accompanying analysis package also retains the original scatter plots and the correlations for the other BTW grid widths.

4.5 What the power-law fits establish

The fits in Equation (3.2) were calculated at $s_{\min} = 5, 10, 20, 50, 100, 200$. Figure 11 shows the two $L = 32$ models. For BTW, the fitted exponent rises from 1.4141 at threshold 5 to 2.4372 at threshold 200. The random model also shows threshold dependence. There is no stable estimate across this full range of lower cutoffs.

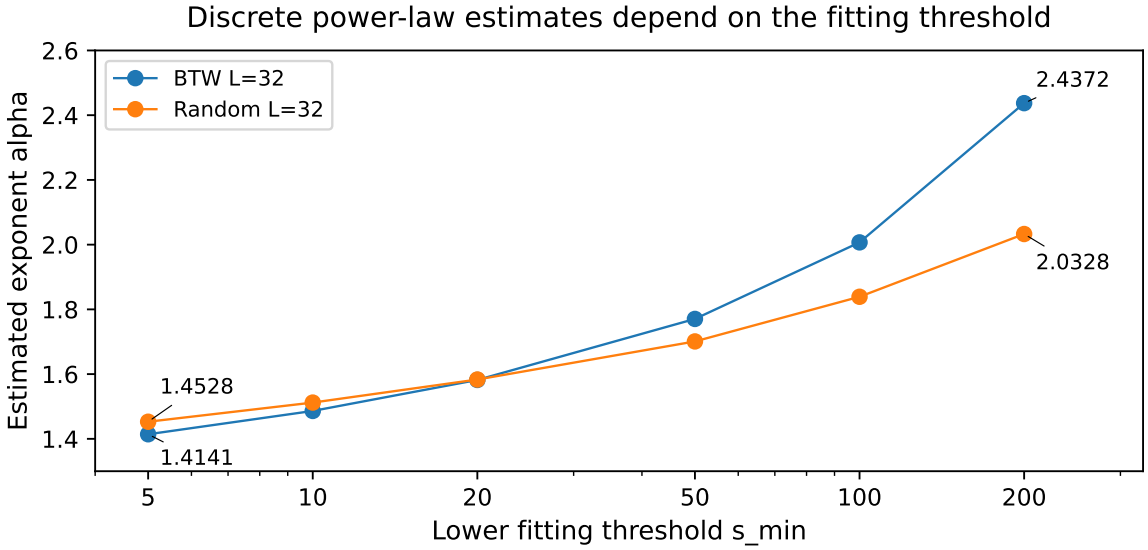


Figure 11: Exploratory discrete maximum-likelihood estimates for the two $L = 32$ settings. Points use the same six thresholds. The labels identify the endpoint estimates; the line segments simply join the estimates. No confidence intervals are claimed.

Table 7: Fits at $s_{\min} = 20$ for all four main settings. KS is the maximum CDF distance and has no p-value interpretation here. All 24 fitted rows are supplied in the data package.

Setting	Tail observations	$\widehat{\alpha}$	KS distance
BTW $L = 16$	8,776	1.9746	0.1319
BTW $L = 32$	12,173	1.5819	0.1349
BTW $L = 64$	14,345	1.4170	0.1322
Random $L = 32$	9,390	1.5833	0.0864

The finite-grid CCDFs bend towards their largest events, so progressively raising the lower threshold changes the portion of the distribution being fitted. An unbounded power law cannot represent every aspect of a finite-grid distribution. Known difficulties with simple finite-size scaling in the Abelian sandpile provide further reason to interpret a single fitted slope cautiously [7].

The present calculation estimates parameters and records a discrepancy measure. It provides no bootstrap goodness-of-fit probability and no likelihood-ratio comparison against a truncated power law or a lognormal model. I therefore retain the visible distribution shapes and the threshold dependence as the main results of this part of the project.

5 Random redistribution extension

5.1 Changing the outgoing directions

The extension keeps the threshold at four. At each toppling, four grains leave the source, and each independently chooses up, down, left or right with probability $1/4$. Several may reach the same neighbour. A direction outside the grid remains a loss event; it is not redrawn. The expected transfer in each direction is still one grain, although its realised value fluctuates.

The comparison uses $L = 32$ and three repeats per rule. Within a repeat, both models receive exactly the same recorded addition positions. Burn-in positions, recorded positions and random outgoing directions have separate random streams. The direction stream continues through burn-in. Thus the doubled-burn-in comparison preserves the later drive positions while allowing a different internal random history.

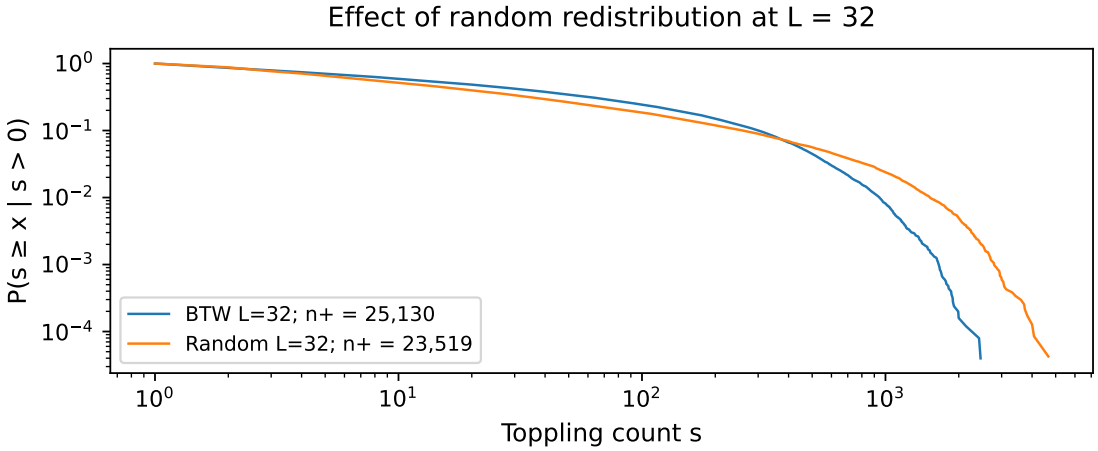


Figure 12: Positive toppling-count distributions at $L = 32$, pooling the three runs per rule. The curves cross, so their ordering depends on the threshold examined. Table 8 reports specific run-level means and quantiles, with their conditioning stated explicitly.

Table 8: Comparison at $L = 32$. Each entry is the mean ± sample SD across three runs. Probabilities use the 0–1 scale. "Event" means an addition with $s > 0$.

Quantity	BTW	Random redistribution
Mean height H	2.0851 ± 0.0011	1.9948 ± 0.0011
$P(s > 0)$, all additions	0.41883 ± 0.00201	0.39198 ± 0.00255
$P(d > 0)$, all additions	0.19850 ± 0.00143	0.15207 ± 0.00093
Mean s, all additions	40.5030 ± 0.2150	40.6439 ± 0.2192
Mean s, event	96.705 ± 0.530	103.688 ± 0.127
Mean a, event	78.294 ± 0.624	50.042 ± 0.475
Mean r, event	7.366 ± 0.037	5.629 ± 0.069
Mean d, all additions	0.999783 ± 0.000957	0.999667 ± 0.000388
99th percentile of s, event	942.57 ± 11.17	1511.24 ± 40.53

The random model triggers fewer avalanches, with a decrease of 2.685 percentage points in event probability. Among triggered events, it has more topplings on average and smaller average spatial extent. The higher 99th percentile in Table 8 describes a particular upper-tail summary, consistent with the differing curve shapes in Figure 12.

5.2 Geometry and average transport

Figure 13 expresses selected differences from Table 8 as relative changes. The conditional mean toppling count increases by 7.22%, while mean area falls by 36.08% and mean radius by 23.57%. This combination is consistent with more repeated toppling within a smaller region. The ratio of the conditional mean s to the conditional mean a changes from about 1.235 to 2.072. It is a ratio of two means; an average of the event-by-event ratio would answer a different question. Individual grain paths were not stored, so this interpretation is based on the measured counts and geometry.

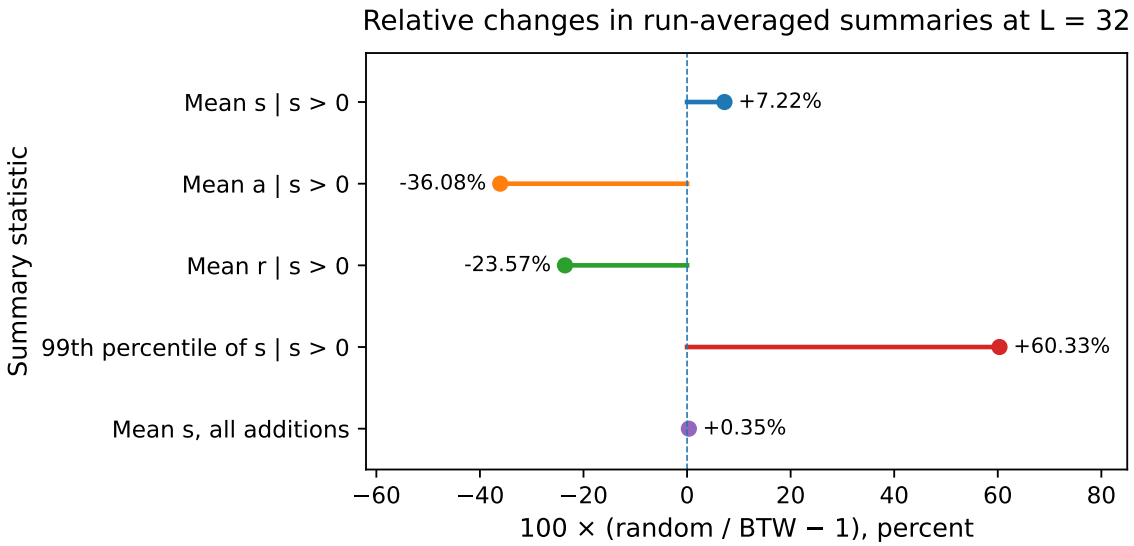


Figure 13: Relative changes from BTW to random redistribution, computed from the unrounded three-run means: $100(\text{random}/\text{BTW} - 1)$. The top three rows and the quantile row condition on $s > 0$. The bottom row includes every addition. These are descriptive changes without confidence intervals.

The all-addition mean toppling counts change little. This has a useful independent check. Set $A = -\Delta$ and let $v(x)$ be the stationary expected number of topplings at x per addition. Uniform driving supplies $1/L^2$ grains per site on average. Expected mass balance gives

$$Av = L^{-2}\mathbf{1}, \qquad \mathbb{E}[s] = \sum_x v(x). \tag{5.1}$$

The random rule has the same expected outgoing transfer, so it gives the same equation under stationarity and finite expected toppling counts. I solved it using $A = I \otimes T + T \otimes I$, where $T = \text{tridiag}(-1, 2, -1)$. The largest residual was below 2.4×10^{-16}.

Table 9: Independent transport calculation and measured all-addition mean topplings. Differences use unrounded values.

Setting	Equation (5.1)	Measured	Relative difference
BTW $L = 16$	11.338215	11.333050	−0.0456%
BTW $L = 32$	40.580318	40.502967	−0.1906%
Random $L = 32$	40.580318	40.643933	+0.1568%
BTW $L = 64$	153.043100	153.140617	+0.0637%

All four measured means lie within 0.2% of the corresponding value. Lower event frequency can coexist with a larger conditional event size. Similarly, fewer loss events can remove more grains each while the mean loss per addition remains near one.

5.3 Sensitivity to burn-in and finite sampling

A burn-in formula alone cannot establish adequate sampling. I checked three windows from the latter half of burn-in, ten windows from the recorded phase, and the first and second halves of each main record. Both $L = 32$ rules were also run with 20,480 burn-in additions, twice the original value, using three paired driving sequences.

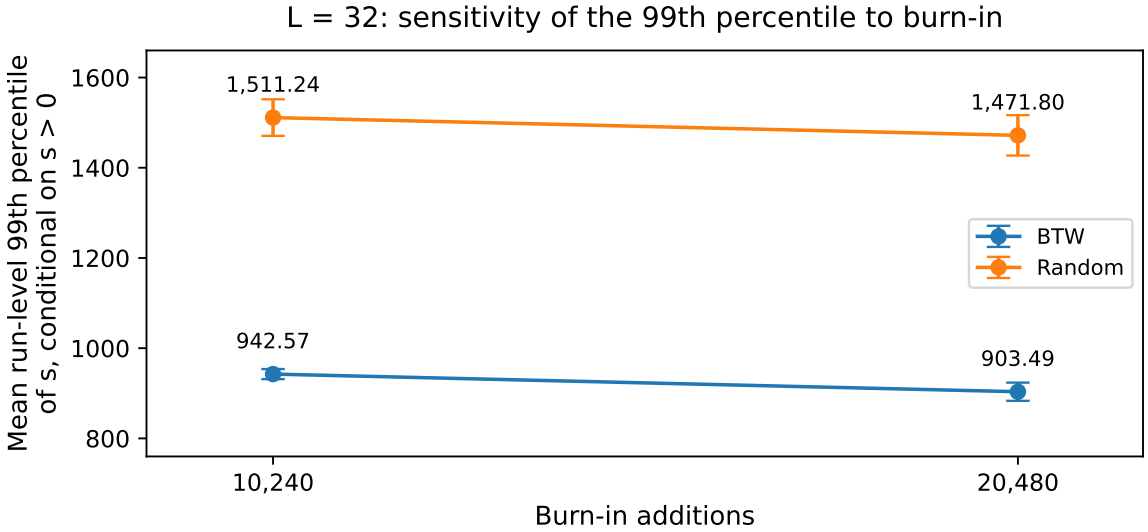


Figure 14: Mean run-level 99th percentiles at the two burn-in lengths, conditional on $s > 0$. Labels show the means to two decimal places and bars show sample SD across three runs. The comparison direction remains the same at both lengths.

Table 10: Sensitivity at $L = 32$. Relative changes use the unrounded estimates. A relative probability change is different from a percentage-point difference.

Rule	Quantity	B = 10,240	B = 20,480	Change
BTW	Mean H	2.085122	2.082551	−0.123%
BTW	$P(s > 0)$	0.418833	0.428083	+2.209%
BTW	99th percentile s	942.57	903.49	−4.146%
Random	Mean H	1.994755	1.993763	−0.050%
Random	$P(s > 0)$	0.391983	0.393000	+0.259%
Random	99th percentile s	1511.24	1471.80	−2.610%

Mean height changes by a small fraction of a percent, but the upper-tail quantiles move by several percent. The BTW 99th percentile falls by 4.146%, and the random-rule value falls by 2.610% (Figure 14). For event probability, the BTW difference is 0.925 percentage points. These shifts can reflect both burn-in history and finite-window variation; they do not isolate a single source of bias.

At both burn-in lengths the random model retains a lower average height, smaller average event area and higher 99th percentile of toppling count. Those directions are better supported than the last reported decimal of a tail estimate. Three-run standard deviations describe variation within the present experiment and leave uncertainty about longer-timescale dependence and rarer events.

Summing Equation (3.1) gives an exact finite-window check,

$$\overline{d}_{\text{all}} = 1 - \frac{M_{\text{end}} - M_{\text{start}}}{N}. \tag{5.2}$$

This explains why the all-addition means in Table 8 are close to one. A small net mass difference over the window is useful evidence for balance, though it cannot by itself establish that every avalanche statistic has converged.

6 Discussion and conclusions

6.1 What changed when redistribution became random

The size comparison and the rule comparison answer different parts of the project. In BTW, increasing grid width increases the available spatial extent and extends the observed avalanche tail. Randomising the outgoing directions at a fixed width changes how activity is arranged within that space. The extension's events have greater conditional toppling counts on average, while their mean area and radius are smaller. The separation of these measurements is important: counting every toppling and counting each visited site once describe different aspects of the same event.

The almost unchanged all-addition mean toppling count is particularly informative. The linear transport calculation depends on the expected transfer to each direction, which is shared by the two rules. It agrees closely with both measured averages. Meanwhile, the probability of triggering an avalanche and the shape of its conditional size distribution change. The numerical results therefore preserve a common first-moment constraint while showing differences in the event-level statistics.

The loss statistics have a similar structure. Many additions cause no loss, and that fraction rises with BTW grid width in these records. When loss does occur, an event can remove several grains. The complete driving sequence balances these intermittent outputs against the steady one-grain input. Conditioning on loss-producing events would change the average and should be stated whenever that sample is used.

6.2 Limitations and possible next steps

The main limitation is the amount of tail information. There are three runs per setting, and large events occupy a small part of the records. The burn-in comparison moves the 99th percentile by several percent. The fitted exponents also change substantially with the lower threshold. Those observations make the report's numerical comparisons useful at the chosen sizes, while limiting any interpretation as a precise measurement of an asymptotic critical exponent.

The stochastic extension was studied at one width. Additional sizes would help separate rule-dependent changes from finite-size effects. Longer independent runs would also improve the upper-tail estimates. A later fitting study could compare a truncated power law and a lognormal distribution, followed by an appropriate goodness-of-fit procedure. These tasks require further calculations; the present fits provide the starting evidence and show where extra sampling would matter.

Processing order was checked path by path only for deterministic BTW. With random redistribution, changing the order while consuming a single global random stream assigns different instructions to different sites. The resulting paths need not match. A separate treatment would be needed to compare fixed random instructions across processing orders.

6.3 Conclusion

The simulations reproduce the expected loading and broad avalanche statistics of the finite open BTW model. The initial equal-height guess fails, and the mean height moves towards the square-lattice benchmark as the grid grows. The deterministic stabilisation has a unique outcome, supported here by an explicit argument and independent numerical implementations.

The main result of the extension is a shift towards more repeated activity within a smaller average spatial region at $L = 32$. This occurs alongside a lower avalanche frequency and a nearly unchanged all-addition mean toppling count. The conserved average transport provides a useful explanation for the latter. The project also shows why sample definitions, processing-order tests and sensitivity checks are needed to make even a short simulation study interpretable.

References

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Acknowledgements and use of assistance

The course handout and supplied Python skeleton were used as model references. AI assistance was used in developing and checking the implementation, organising the analysis, and drafting and editing this report. The calculations are linked to the saved simulation data supplied with the project.

Data and code availability

The companion archive `1873749_Software.zip`, provided for the Project software submission, includes the 300-line main program, its pinned dependency list, all 38 saved NumPy run files, and the tables and figures used here. Additional reporting scripts rebuild the figure assets and the numerical checks from those records. They are separate from the main simulation program. Appendix A gives the question coverage and basic run commands.

A Course question coverage and reproducibility

Connection to the handout

The report follows a continuous account of the project. Table 11 identifies the location of each course question and the selected extension.

Table 11: Coverage of the original sandpile questions.

Question	Topic	Report location
1	Scale invariance and power laws	Section 1.2
2	Conditions relevant to self-organised criticality	Chapter 1, Section 4.1
3	Open boundaries and the infinite-grid question	Section 1.3
4	Initial mean-height prediction and measured result	Sections 4.1 and 3.3
5	Stable configurations, stationarity and equilibrium	Section 4.1
6	Time sampling, ensembles and ergodicity	Sections 2.3 and 4.2
7	Topplings, area, loss, length and distribution fits	Sections 4.3–4.5
8	Predicted and measured correlations	Section 4.4
9	Meaning of the matrix conditions	Section 2.2
10	BTW matrix and threshold	Sections 2.1 and 2.2
11	Exchange of two legal topplings	Section 2.3
12	Termination and unique stabilisation	Section 2.3
Extension 2	Independent random directions for four grains	Chapter 5

Running the software

The reference environment used Python 3.13.5, NumPy 2.3.5, SciPy 1.17.0 and Matplotlib 3.10.8. In the supplied `software` directory, the commands are

```
python -m pip install -r requirements.txt
python sandpile.py test --out check_results
python sandpile.py demo --out demo_results
python sandpile.py full --out reproduced_results
python sandpile.py analyse --out results
```

The full command runs the 38 experiments. Use a new output directory to keep it separate from the supplied reference records. The analyse command rebuilds the core summaries and original figure set from saved data; the separate reporting script creates the figure set in this report. Do not use Python's optimisation flag for the tests, because it disables assertions.

Each NPZ file stores the full trajectory, including burn-in, addition positions, final height grid, the largest recorded event's toppling-count array, and JSON metadata. The main seeds are 1000–1002; the independent ensemble uses 2000–2019. Metadata records each run's size, burn-in, sampling length and redistribution rule. Hashes and recalculated numerical tables are included under `verification`.

A small correction to the handout example

Applying the rule to the unstable grid $\begin{pmatrix} 1 & 3 & 3 \\ 2 & 4 & 2 \\ 2 & 1 & 1 \end{pmatrix}$ gives $\begin{pmatrix} 2 & 1 & 1 \\ 3 & 2 & 0 \\ 2 & 2 & 2 \end{pmatrix}$, with four topplings and four grains lost. Its mass changes from 19 to 15. No extra grain is added in this relaxation test. The printed diagram misses a returning grain at the centre; the program checks the stated transfer rule and exact mass balance.